not a homework

P397-习题 13.2　A 类　2-1, 2-5, 2-9; 14-1, 14-3, 14-5
P407-习题 13.3　A 类　2-2, 2-3; 5-1; 6

2-1、计算 $\int_0^1 \frac{\arcsin x}{\sqrt{1-x^2}}\,dx = \frac{1}{2}\arcsin^2 x\Big|_0^1 = \frac{\pi^2}{4}$

2-5、计算 $\int_0^3 \frac{dx}{\sqrt[3]{3x-1}} = (3x-1)^{\frac{2}{3}}\frac{1}{2}\Big|_0^3$

2-9、计算 $\int_0^2 \frac{dx}{\sqrt{|x^2-1|}}$

$$I = \int_0^1 \frac{dx}{\sqrt{1-x^2}} + \int_1^2 \frac{dx}{\sqrt{x^2-1}}$$

$$= \arcsin x\Big|_0^1 + \ln\left(x+\sqrt{x^2-1}\right)\Big|_1^2 = \frac{\pi}{2} + \ln(2+\sqrt{3})$$

14-1、判断敛散性 $\int_a^b \frac{dx}{\sqrt{(x-a)(b-x)}}, (b>a)$

解: 当 $x \to a+$ 时 $\frac{1}{\sqrt{(x-a)(b-x)}} \sim \frac{(b-a)^{-1/2}}{\sqrt{x-a}}$ 故 $\int_a^{\frac{a+b}{2}} \frac{dx}{\sqrt{(x-a)(b-x)}}$ 收敛

当 $x \to b-$ 时 $\frac{1}{\sqrt{(x-a)(b-x)}} \sim \frac{(b-x)^{-1/2}}{\sqrt{b-a}}$ 故 $\int_{\frac{a+b}{2}}^b \frac{dx}{\sqrt{(x-a)(b-x)}}$ 收敛

由此 $\int_a^b \frac{dx}{\sqrt{(x-a)(b-x)}}$ 收敛

14-3、判断敛散性 $\int_0^1 \frac{dx}{\sqrt[4]{1-x^4}}$

解: $x \to 1-$ 时,

$$(x-1+1)^4 - 1 \sim 4(x-1) \Rightarrow \frac{1}{\sqrt[4]{1-x^4}} \sim \frac{1}{\sqrt{2}}(1-x)^{-1/4} \Rightarrow \text{积分收敛}$$

14-5、判断敛散性 $\int_0^2 \frac{e^{-2x}\,dx}{(2-x)^2}$

解: $x \to 2-$ 时,

$$\frac{e^{-2x}\,dx}{(2-x)^2} \sim \frac{e^{-4}}{(2-x)^2} \Rightarrow \text{积分发散}$$

2-2、求导数 $\varphi(x)=\int_0^x \frac{\ln(1+xy)}{y}\,dy$

$$\varphi'(x)=\frac{\ln(1+x^2)}{x}+\int_0^x \frac{dy}{1+xy}=\frac{\ln(1+x^2)}{x}+\frac{\ln(1+xy)}{y}\Big|_0^x=2\frac{\ln(1+x^2)}{x}$$

2-3、求导数 $\varphi(x)=\int_{x^2}^{x^3} \arctan\frac{y}{x}\,dy$

$$\begin{aligned}\varphi'(x)&=3x^3\arctan x^2-2x\arctan x+\int_{x^2}^{x^3}\frac{dy}{1+\frac{y^2}{x^2}}\frac{-y}{x^2}\\&=\ldots+\int_{x^2}^{x^3}\frac{-y\,dy}{x^2+y^2}=\ldots+\frac{-1}{2}\ln(x^2+y^2)\Big|_{x^2}^{x^3}\\&=3x^3\arctan x^2-2x\arctan x+\frac{1}{2}\ln\frac{2}{1+x^4}\end{aligned}$$

5-1、计算积分 $I=\int_0^1 \frac{\arctan x}{x}\frac{dx}{\sqrt{1-x^2}}$

解: 记 $f(\theta)=\int_0^1 \frac{\arctan \theta x}{x}\frac{dx}{\sqrt{1-x^2}}$, 则

$$f'(\theta)=\int_0^1 \frac{dx}{\sqrt{1-x^2}(1+\theta^2x^2)}=(x=\sin\ t)=\int_0^{\frac{\pi}{2}} \frac{dt}{1+\theta^2\sin^2 t}$$

$$=(\tan t=u)=\int_0^{\infty} \frac{du/(1+u^2)}{1+\theta^2\frac{u^2}{1+u^2}}=\int_0^{\infty} \frac{du}{u^2+1+\theta^2}$$

$$=\frac{1}{\sqrt{1+\theta^2}}\arctan\frac{u}{\sqrt{1+\theta^2}}\Big|_0^{+\infty}=\frac{1}{\sqrt{1+\theta^2}}$$

$$\Rightarrow f(\theta)=\ln(\theta+\sqrt{1+\theta^2})+c\ \ f(0)=0\Rightarrow f(\theta)=\ln(\theta+\sqrt{1+\theta^2})\Rightarrow I=\ln(1+\sqrt{2})$$

6、假设 f 连续, 证明 $\lim\limits_{h\to 0+}\dfrac{1}{h}\displaystyle\int_a^x \big(f(t+h)-f(t)\big)\,dt = f(x)-f(a)$

证: (应用积分中值定理)

$$\text{由}\quad \int_a^x \big(f(t+h)-f(t)\big)\,dt = \int_a^x f(t+h)\,dt - \int_a^x f(t)\,dt$$

$$= \int_{a+h}^{x+h} f(t)\,dt - \int_a^x f(t)\,dt = \int_x^{x+h} f(t)\,dt - \int_a^{a+h} f(t)\,dt$$

$$\Rightarrow \frac{1}{h}\int_a^x \big(f(t+h)-f(t)\big)\,dt = f(\theta_1)-f(\theta_2)\ \text{ 其中 } \theta_1\in(x,x+h),\theta_2\in(a,a+h)$$

$$\Rightarrow \lim_{h\to 0+}\frac{1}{h}\int_a^x \big(f(t+h)-f(t)\big)\,dt = f(x)-f(a)$$